

Ex 7.2
Q1 If A is an orthogonal matrix show that $A = 1$ or -1

Sol Since A is orthogonal Matrix

$$\therefore AA^t = I$$

$$\det(AA^t) = \det I$$

By Product Theorem

$$\Rightarrow \det(A) \cdot \det(A^t) = 1$$

$$\therefore (\det(A) = \det A^t \cdot \det B)$$

$$\Rightarrow \det(A) \cdot \det(A) = 1$$

$$\therefore \det(A^t) = \det(A)$$

$$\Rightarrow [\det(A)]^2 = 1$$

$$\det(A) = \pm 1 \text{ Proved}$$

2nd Method

Since A is orthogonal Matrix

$$\therefore A^t = A^{-1}$$

$$\det(A^t) = \det(A^{-1})$$

$$\det A = \frac{1}{\det(A)}$$

$$\therefore \det(A^t) = \det A$$

$$[\det A]^2 = 1$$

$$\det A = \pm 1 \text{ Proved.}$$

Prove that $\det A^t = \frac{1}{\det A}$

$$\text{We know } A^t A = I$$

$$\det(A^t A) = \det I$$

$$\text{By Product Theorem } \det A^t \det A = 1$$

$$\det A^t = \frac{1}{\det A} \text{ proved.}$$

Q2 Find an orthogonal matrix whose first row is $(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$

Sol Let $R_1 = (\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$ (given)

A vector orthogonal to R_1 is

$$R_2^* = \begin{bmatrix} e_1 & e_2 \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}$$

$$= \frac{2}{\sqrt{5}} e_1 - \frac{1}{\sqrt{5}} e_2$$

$$= (\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}})$$

$$\|R_2^*\| = \sqrt{\frac{4}{5} + \frac{1}{5}} = \sqrt{\frac{5}{5}} = 1$$

$$R_2 = \frac{R_2^*}{\|R_2^*\|} = (\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}) = (\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}})$$

$$\therefore A = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \end{bmatrix}$$

To Check $AA^t = I$

$$AA^t = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \frac{1}{5} + \frac{4}{5} & \frac{2}{5} - \frac{2}{5} \\ \frac{2}{5} + \frac{2}{5} & \frac{4}{5} + \frac{1}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ proved}$$

Q3 Find an orthogonal matrix whose first row is multiple of (1,1,1)

Sol $\omega_1 = (1, 1, 1)$ given

$$\|\omega_1\| = \sqrt{3}$$

Normalised first Row $V_1 = \frac{\omega_1}{\|\omega_1\|} = \frac{(1, 1, 1)}{\sqrt{3}} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

Let ω_2 is orthogonal to V_1 , where $\omega_2 = (x, y, z)$

$$\therefore \langle V_1, \omega_2 \rangle = 0$$

$$\Rightarrow \frac{x}{\sqrt{3}} + \frac{y}{\sqrt{3}} + \frac{z}{\sqrt{3}} = 0$$

$$\Rightarrow x + y + z = 0$$

Put $x = 0$

$$\Rightarrow y + z = 0 \Rightarrow z = -y$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ y \\ -y \end{bmatrix} = y \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$\therefore \omega_2 = (0, 1, -1)$$

$$\|\omega_2\| = \sqrt{0^2 + 1^2 + (-1)^2} = \sqrt{2}$$

$$V_2 = \frac{\omega_2}{\|\omega_2\|} = \frac{(0, 1, -1)}{\sqrt{2}} = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

A vector orthogonal to V_1 & V_2 is

$$\omega_3 = \begin{vmatrix} e_1 & e_2 & e_3 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{vmatrix}$$

$$= e_1 \left(-\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}}\right) - e_2 \left(-\frac{1}{\sqrt{6}}\right) + e_3 \left(\frac{1}{\sqrt{6}}\right)$$

$$\omega_3 = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$\|\omega_3\| = \sqrt{\frac{4}{6} + \frac{1}{6} + \frac{1}{6}} = 1$$

$$V_3 = \frac{\omega_3}{\|\omega_3\|} = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

Hence the required orthogonal matrix is $\begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix}$

we cannot use

$$\begin{vmatrix} e_1 & e_2 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix} + \begin{vmatrix} e_2 & e_3 \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix} + \begin{vmatrix} e_3 & e_1 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix}$$

because it becomes 0, and when it becomes 0 it is not applicable.

Q4 Find an orthogonal matrix whose first row is $(0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$

$$\text{Let } R_1 = \left(0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$$

A vector (row) orthogonal to R_1 is $R_2^* = \begin{vmatrix} e_1 & e_2 \\ 0 & \frac{1}{\sqrt{5}} \end{vmatrix} + \begin{vmatrix} e_2 & e_3 \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{vmatrix} + \begin{vmatrix} e_3 & e_1 \\ \frac{2}{\sqrt{5}} & 0 \end{vmatrix}$

$$= e_1\left(\frac{1}{\sqrt{5}}\right) + \frac{2}{\sqrt{5}}e_2 - \frac{1}{\sqrt{5}}e_3 + (0 - \frac{2}{\sqrt{5}}e_1)$$

$$= e_1\left(\frac{1}{\sqrt{5}} - \frac{2}{\sqrt{5}}\right) + \frac{2}{\sqrt{5}}e_2 - \frac{1}{\sqrt{5}}e_3$$

$$R_2^* = -\frac{1}{\sqrt{5}}e_1 + \frac{2}{\sqrt{5}}e_2 - \frac{1}{\sqrt{5}}e_3 = \left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right)$$

$$\|R_2^*\| = \sqrt{\frac{1}{5} + \frac{4}{5} + \frac{1}{5}} = \sqrt{\frac{6}{5}}$$

$$R_2 = \frac{R_2^*}{\|R_2^*\|} = \frac{\left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right)}{\sqrt{\frac{6}{5}}} = \frac{\sqrt{5}}{\sqrt{6}} \left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right) = \left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right)$$

A vector orthogonal to R_1 & R_2 is R_3^*

$$R_3^* = \begin{vmatrix} e_1 & e_2 & e_3 \\ 0 & \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{vmatrix}$$

$$= e_1\left(-\frac{1}{\sqrt{30}} - \frac{4}{\sqrt{30}}\right) - e_2\left(\frac{2}{\sqrt{30}}\right) - e_3\left(\frac{1}{\sqrt{30}}\right)$$

$$R_3^* = \left(-\frac{5}{\sqrt{30}}, -\frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}}\right)$$

$$\|R_3^*\| = \sqrt{\frac{25}{30} + \frac{4}{30} + \frac{1}{30}} = 1$$

$$R_3 = \frac{R_3^*}{\|R_3^*\|} = \left(-\frac{5}{\sqrt{30}}, -\frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}}\right)$$

Hence orthogonal Matrix $A = \begin{pmatrix} 0 & \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ -\frac{5}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{1}{\sqrt{30}} \end{pmatrix}$

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Q5 Show that the products and inverses of orthogonal matrices are orthogonal. Hence show that orthogonal matrices form a group under multiplication.

Sol Let A & B be orthogonal matrices. So $A^t A = I = A^t = A^{-1}$
 $B^t B = I \Rightarrow B^t = B^{-1}$

Now To Prove Product of orthogonal Matrix is Orthogonal, i.e. (AB) is orthogonal.

Therefore $(AB)^t (AB) = B^t A^t AB$ we show $(AB)^t (AB) = I$
 $= B^t (A^t A) B$
 $= B^t I B$
 $= B^t B$
 $(AB)^t (AB) = I$

So AB is an orthogonal Matrix.

Now To Prove Inverse of orthogonal Matrix is orthogonal, i.e. A^{-1} is orthogonal.

Therefore $(A^{-1})^t (A^{-1}) = (A^{-1})^t (A^t)$ we show $(A^{-1})^t (A^{-1}) = I$
 $= (A^t)^t (A^t) \quad (\because (A^{-1})^t = A^t \text{ since } A \text{ is orthogonal})$
 $= (AA^t)^t$
 $= (I)^t$
 $(A^{-1})^t (A^{-1}) = I$

Hence A^{-1} is orthogonal.

To Prove Set of orthogonal matrices form a group under multiplication.

Let $\mathcal{G}$ = Set of orthogonal matrices.

i) Let $A, B \in \mathcal{G}$ then $AB \in \mathcal{G}$ $\because$ product of orthogonal is orthogonal.

Hence $\mathcal{G}$ is closed under B.O of multiplication.

ii) In Matrices Associative law holds.

iii) Identity Matrix is always orthogonal Matrix. So $I \in \mathcal{G}$.

iv) Inverse of each orthogonal matrix is orthogonal (as solved above).

So Inverse of each orthogonal matrix belonging to $\mathcal{G}$ is in $\mathcal{G}$.

So Inverse exists in $\mathcal{G}$.

All conditions are satisfied, Hence $\mathcal{G}$ is a group under multiplication.